DLX Architecture Overview

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ECE 6930 – Advanced Reconfigurable Computing

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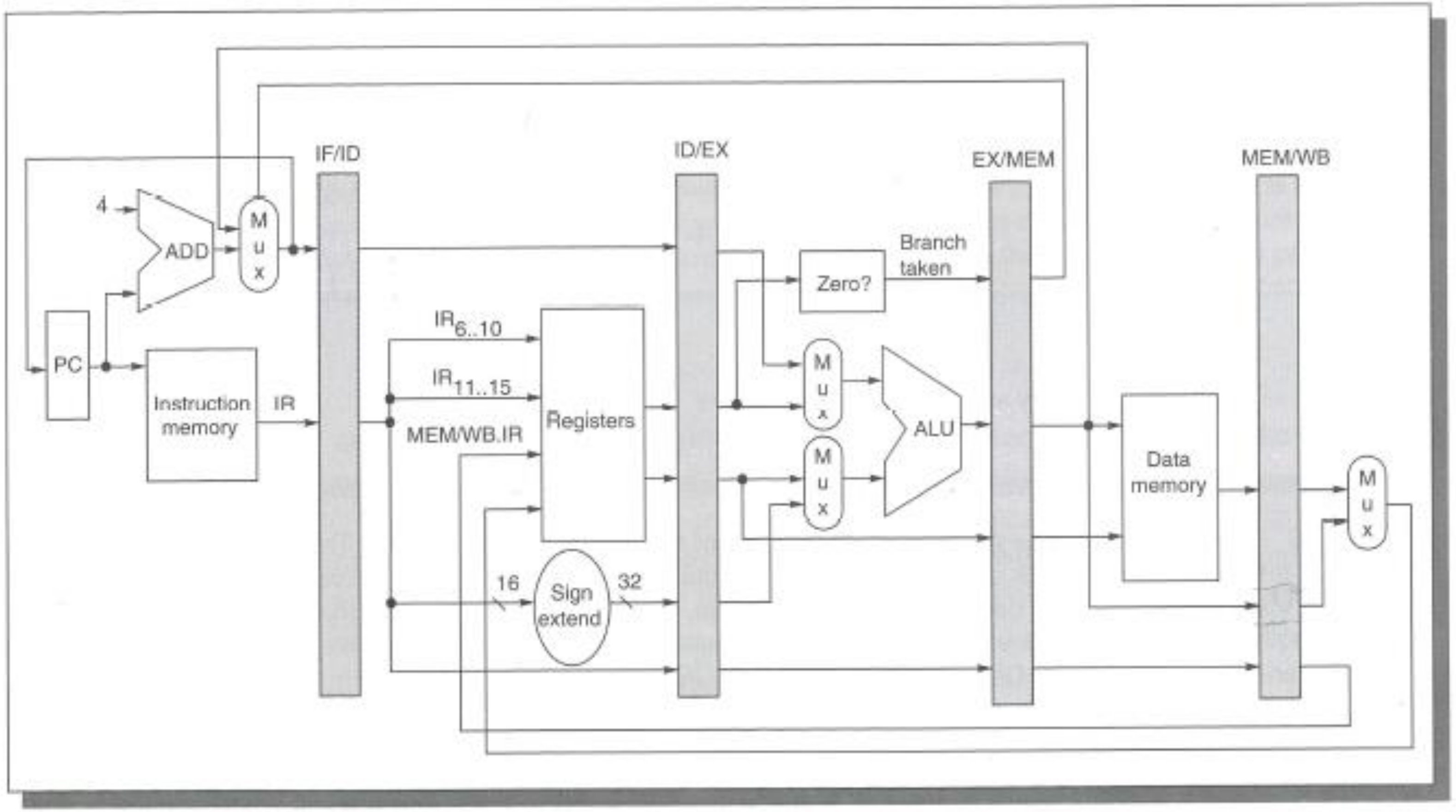
- Remember that this is a Harvard Architecture – Instruction Memory and Data Memory are separate.

DLX Pipeline Architecture

- DLX consists of a 5 stage pipeline
  - Fetch – Get the next instruction out of the instruction memory
  - Decode – Make sense of the opcode and identify the operands
  - Execute – Perform the operation on the operands
  - Memory – Access memory (if needed)
  - Write-Back – Store result of operation in register (if needed)

DLX Pipeline Architecture



VHDL Packages

- https://www.nandland.com/vhdl/examples/example-package.html